Theoretical and Applied Climatology (2021) 145:1243–1258
https://doi.org/10.1007/s00704-021-03700-2

ORIGINAL PAPER

Evaluating the impact of projected climate on rice–wheat-groundnut cropping sequence in lower Gangetic plains of India: a study using multiple GCMs, DSSAT model, and long-term sequence analysis

M. A. Sarath Chandran[1,2] · Saon Banerjee[1] · Asis Mukherjee[1] · Manoj K. Nanda[1] · Subhasis Mondal[3] · V. Visha Kumari[2,4]

Received: 21 October 2020 / Accepted: 14 June 2021 / Published online: 23 June 2021
© The Author(s), under exclusive licence to Springer-Verlag GmbH Austria, part of Springer Nature 2021

Abstract
Cropping intensification is a solution to ensure global food security amid the threats of climate change and variability. Crop models are used extensively to evaluate the performance of single crop for a particular year under future climate. However, very few literatures are available on the application of crop models to evaluate intensive cropping systems. The study attempts to address this issue by evaluating the impact of projected climate derived from 29 GCMs during four future scenarios (mid and end-century under RCP4.5 and RCP8.5) on the performance of rice–wheat-groundnut cropping sequence in Mohanpur, India, using the sequence analysis module of DSSAT. A marked reduction in crop duration was simulated for rice (4 to 17 days) and wheat (1 to 16 days) during all four future scenarios. On the contrary, the duration of groundnut increased by 1–4 days. Under elevated CO_2 concentration, an increase in biomass was simulated for all three crops. During future periods under elevated CO_2, the yield of rice tends to increase under all the future scenarios whereas wheat and groundnut could decrease the magnitude of yield reduction due to an increase in temperature. Rice was benefited due to the carry-over effect of residue from preceding groundnut and, hence, could sustain the yield on the long-term. The uncertainty in simulation due to the selection of GCMs was also quantified. The study provided a better understanding on how a cropping sequence responds to projected climate rather than a single crop in a sequence, which can be used to develop better adaptation strategies.

1 Introduction

Crop production is dependent on the performance of cultivated varieties, which are adapted to a set of environmental conditions. Climate, with its spatial and temporal variability, is a major influencing factor of crop production. Thus, any change in climatic elements is bound to have either positive or negative impacts on crop production. Climate change can alter the rainfall distribution and soil water balance, may reduce crop duration, and increase the yield variability and change the areas suitable for growing many crops (Olesen and Bindi, 2002). Over the years, the challenge for crop research scientists is to produce more food from less water, land, and nutrients to meet the demand of an ever-increasing population. But, along with this, the carbon footprint of crop production needs to be reduced. While an increase in temperature during the crop-growing season reduces the yield, CO_2 fertilization effect to an extent can reduce its negative impact. So, there is a trade-off between all these factors to ensure food security under projected climate. Crop simulation modeling offers a research tool for evaluating trade-offs of these potential challenges (Mathews et al., 2013).

The Decision Support System for Agrotechnology Transfer (DSSAT) model has been widely used around the world to assess the impact of future climate on the growth and yield of crops/cropping systems and for identification of possible adaptation strategies required to address the negative impacts (Tsuji et al., 1994; Jones et al., 2003; Hoogenboom et al., 2004). The models included in DSSAT v4.7 (CERES

✉ M. A. Sarath Chandran
sarathagri@gmail.com

[1] Department of Agricultural Meteorology & Physics, Bidhan Chandra Krishi Viswavidyalaya, Mohanpur 741 252, West Bengal, India

[2] ICAR-Central Research Institute for Dryland Agriculture, Santoshnagar, Hyderabad 500 059, Telangana, India

[3] Department of Plant Physiology, Bidhan Chandra Krishi Viswavidyalaya, Mohanpur 741 252, West Bengal, India

[4] Department of Agronomy, Bidhan Chandra Krishi Viswavidyalaya, Mohanpur 741 252, West Bengal, India

for cereals, CROPGRO for pulses) are process-oriented and dynamic models that simulate growth, development, and yield for more than 32 crops. The Century model (Parton et al., 1993) integrated into DSSAT simulates the long-term dynamics of C for different plant–soil systems. It was developed to analyze a wide range of crop rotations and tillage practices and to determine the impact of crop management on the productivity and sustainability of agroecosystems.

Most of the studies on the impact of climate change in crop production using crop models are confined to simulating the performance of a single crop for a single year (Palosuo et al., 2011; Rötter et al., 2012; Asseng et al., 2013; Bassu et al., 2014; Banerjee et al., 2016). In such studies, the initial condition of soil (water, organic matter, and nutrients) was kept constant at the onset of each growing season. Along with this, the carry-over effect from previous crops such as N mineralizing from the harvest residues or the changes in soil water content during the non-cropping period due to evaporation is ignored (Ewert et al., 2015).

Crop rotations are considered an option for increasing the resilience of agricultural system (Smith et al., 2008; Reidsma et al., 2009; Lin, 2011) and for developing adaptation strategies to climate change (Olesen et al., 2011). In contrast with the drawbacks of simulation of single year/single crops, Teixeira et al. (2015) reported the superiority of simulating continuous crop rotations compared to single crops and years for a single location in New Zealand. Another model-intercomparison study on European crop rotation reported that continuous simulation of multi-year crop rotations performed better compared to the simulation of single years and single crops (Kollas et al., 2015). Timsina et al. (1997) had used the sequence analysis module of DSSAT to assess the sustainability of the rice–wheat cropping system in the Indo-Gangetic region of India and reported a decrease in rice yields over the years due to decline in organic carbon and nitrogen status.

In new alluvial zone of West Bengal state in India, crop production is taken up in three seasons of a year. During the southwest monsoon season (known as '*kharif*' in India), rice is the major crop, followed by wheat/mustard/maize/lentil/chickpea during post-monsoon season (*Rabi* season). The summer crops comprise of summer rice, vegetables, groundnut, pulse crops like black gram, green gram, etc. This intensive cropping system is supported by an average annual rainfall of 1435 mm and an abundant source of irrigation. This study is aimed at modeling the impact of projected climate on growth and yield of rice–wheat-groundnut cropping sequence in Mohanpur, which comes under new alluvial soil region of West Bengal, India. The specific objectives were (1) to assess the impact of projected climate on performance of rice–wheat-groundnut using long-term (30 years) sequence analysis, (2) to quantify the uncertainty in climate change impact projections using multiple GCMs under various future climate scenarios, and (3) to quantify the impact of elevated CO_2 in mitigating the reduction in yield caused by the projected rise in temperature.

2 Materials and methods

2.1 Study area

The study area consisted of Mohanpur, which comes under the new alluvial zone of West Bengal state, which comes in the lower Gangetic plain of India. The geographical location of the site is 22.56° N and 88.33° E, with an elevation of 10 m above mean sea level. As per the United States Department of Agriculture (USDA) modern taxonomical classification, soil is classified as Aeric Haplaquept. The texture was sandy loam, with moderately well drainage. The physicochemical characteristics of the soil at the study area are presented in Table 1. The climate is typically subtropical—hot and humid.

2.2 Climate scenarios

The observed daily meteorological data for baseline (1980–2010) as well as for the model calibration and validation period (2007–2016) was collected from the Agromet observatory maintained by All India Coordinated Research

Table 1 Physical and chemical properties of the study area used in model evaluation and application

Depth (cm)	LL ($cm^3\ cm^{-3}$)	DUL ($cm^3\ cm^{-3}$)	SAT ($cm^3\ cm^{-3}$)	SRGF	BD ($g\ cm^{-3}$)	SOC %)
0–15	0.078	0.244	0.55	1	1.46	0.58
15–30	0.072	0.23	0.36	0.51	1.59	0.51
30–45	0.072	0.23	0.44	0.31	1.62	0.48
45–60	0.072	0.23	0.43	0.15	1.62	0.47
60–75	0.1	0.23	0.43	0.15	1.61	0.47
75–100	0.1	0.24	0.53	0.15	1.61	0.47
100–150	0.1	0.21	0.53	0.15	1.49	0.47

*LL*lower limit; *DUL* drained upper limit; *SAT* saturation; *SRGF* relative root distribution; *BD* bulk density; *SOC* soil organic carbon

Project on Agrometerology, Mohanpur Centre (BCKV). Future climate scenarios were generated from 29 general circulation models (GCMs). The list of GCMs considered is listed in Table 2.

2.2.1 Subsetting of GCMs

In the present study, 5 GCMs which represented a pool of 29 GCMs were selected using 'Representative Temperature and Precipitation (T&P) GCM Subsetting Approach' (Ruane and McDermid, 2017). More details can be found in supplementary materials. The selected five GCMs, namely, S (MRI-CGCM3), W (CMCC-CMS), K (HadGEM2-ES), O (MIROC5), and H (GFDL-ESM2G) were used for the present study.

Future climate scenarios for the selected five GCMs were generated for two periods viz., mid-century (2040–69) and end-century (2070–99) under representative concentration pathways RCP4.5 and RCP8.5 by applying enhanced-delta technique, which generates mean and variability change scenarios (Ruane et al., 2015). 'R' script was used to generate the future climate as presented in AgMIP (2013). The AgMIP central year CO_2 assumption for mid-century under RCP4.5 (499 ppm) and RCP8.5 (571 ppm) and for end-century under RCP4.5 (532 ppm) and RCP8.5 (801 ppm) were also adopted (AgMIP, 2012).

2.3 Crop simulation model

The Decision Support System for Agrotechnology Transfer (DSSAT) modeling system is a process-based model which is widely used around the world for climate change impact assessment in field crops. It has different modules to simulate water balance, nutrient dynamics, crop growth, phenology, biomass, and yield on a daily basis during the crop growth based on crop characteristics like phenology, photoperiod, leaf area development, and biomass partitioning, which are defined by cultivar-specific genetic coefficients. The input data requirement for the DSSAT includes daily weather data, soil data, crop management data, and observed crop data for model calibration and validation. DSSAT consists of crop-specific models for simulating the growth and yield of various crops. In this study, CERES-Rice, CERES-Wheat, and CROPGRO-Peanut models in DSSAT v4.7 were used for calibration and validation of rice, wheat, and groundnut, respectively.

2.3.1 Model calibration and validation

The crop data required for model calibration and validation were obtained from AICRP on Agrometerology, Mohanpur Centre (2016), Thentu (2016), and Fangzauva (2011) for wet-season rice, wheat, and groundnut, respectively. Experimental data collected were used to calibrate and validate the three models individually so that the optimized parameters during calibration represent the response of crops under study to soil, weather conditions, and management practices in the study region. The details of data used for calibration and validation are presented in Supplementary Table 1.

Model calibration was done by optimizing the values of genetic coefficients of specific cultivars (mentioned in Supplementary Table 1) using the 'Generalized Likelihood Uncertainty Analysis' (GLUE) tool (Jones et al., 2011). Genetic coefficients related to phenology were optimized first so that the difference between simulated and observed phenology was minimum. Then coefficients related to growth, biomass, and yield were optimized for all three crop cultivars.

Table 2 Summary of 29 GCMs used in this study for climate change impact assessment (Ruane and McDermid, 2017)

Symbol	GCM	Horizontal resolution	Symbol	GCM	Horizontal resolution
A	ACCESS1-0	1.25°×1.875°	P	MIROC-ESM	~2.8°×2.8°
B	BCC-CSM 1–1	~2.8°×2.8°	Q	MPI-ESM-LR	~1.9°×1.875°
C	BNU-ESM	~2.8°×2.8°	R	MPI-ESM-MR	~1.9°×1.875°
D	CanESM2	~2.8°×2.8°	S	MRI-CGCM3	~1.1°×1.125°
E	CCSM4	~0.9°×1.25°	T	NorESM1-M	~1.9°×2.5°
F	CESM1-BGC	~0.9°×1.25°	U	FGOALS-g2	~2.8°×2.8°
G	CSIRO-Mk3-6–0	~1.9°×1.875°	V	CMCC-CM	~0.75°×0.75°
H	GFDL-ESM2G	~2.0°×2.5°	W	CMCC-CMS	~1.875°×1.875°
I	GFDL-ESM2M	~2.0°×2.5°	×	CNRM-CM5	~1.4°×1.4°
J	HadGEM2-CC	1.25°×1.875°	Y	HadGEM2-AO	1.25°×1.875°
K	HadGEM2-ES	1.25°×1.875°	Z	IPSL-CM5B-LR	~2.5°×1.26°
L	INM-CM4.0	1.5°×2.0°	1	GFDL-CM3	2°×2.5°
M	IPSL-CM5A-LR	~1.9°×3.75°	2	GISS-E2-R	2°×2.5°
N	IPSL-CM5A-MR	~1.3°×2.5°	3	GISS-E2-H	2°×2.5°
O	MIROC5	~1.4°×1.4°			

For validation, the root mean square error (RMSE), normalized RMSE (nRMSE), and mean absolute error (MAE) were considered. RMSE was calculated using Eq. (1).

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (S_i - O_i)^2}{n}} \quad (1)$$

where n is the total number of observations, S_i is the simulated values and O_i is the observed values. The RMSE gives the standard deviation of the model prediction error. A smaller value indicates better model performance. The nRMSE was calculated using Eq. (2).

$$nRMSE = \left[\frac{RMSE}{\bar{X}}\right] \times 100\% \quad (2)$$

where $\bar{X}$ is the mean of observed values. It gives a measure of the relative difference between observed and simulated values in percentage (Soler et al., 2007). Though RMSE is a popular choice among researchers for the evaluation of model accuracy, nRMSE is considered better than RMSE because it provides the error level associated with each pair of simulated and observed variables (Rahman et al., 2018). Values < 10% are considered excellent prediction while 10 to 20% are good, > 20 to 30% are considered fair, and > 30% indicate poor model performance (Jamieson et al., 1991). MAE was calculated using Eq. (3).

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |S_i - O_i| \quad (3)$$

MAE indicates the absolute difference between simulated and observed variables.

2.4 Sequence analysis

The simulation of cropping sequence requires continuous estimation of soil-related processes like daily soil water balance including the no-crop period between two crops. The carry-over from the preceding crop like crop residue and residual soil moisture should also be made available for the succeeding crop. The sequence analysis module in DSSAT does all the above-mentioned processes and is an ideal tool for simulating the performance of a cropping sequence in long term. The sequence analysis module in DSSAT is based on the following hypotheses.

- Cropping systems that have some sort of problem will eventually exhibit decreasing performance levels over time, be it in terms of yields or yield attributes;
- Some cropping systems will have problems over long periods of time that are exhibited in terms of an increase in variability over the sequence.

In the present study, sequence analysis of rice–wheat-groundnut cropping sequence was performed by running CERES-Rice, CERES-Wheat, and CROPGRO-Peanut crop models in sequence, with initial soil conditions specified only for the first model in the sequence, which is CERES-Rice. Sequence analysis module in DSSAT was used for simulating the cropping system under study for three time periods, viz., 1980–2010 (baseline), 2040–2069 (mid-century), and 2070–2099 (end-century). Furthermore, to understand the effect of elevated CO_2 during future time periods on crop growth and yield, the simulations were undertaken with baseline CO_2 concentration (360 ppm) and elevated CO_2 concentration (e.g. 499 ppm for mid-century-rcp4.5, as mentioned earlier). The details of the crop management data used for the sequence analysis in DSSAT are presented in Supplementary Table 2. The planting date for every year was fixed as presented in Supplementary Table 2. The harvest date for only groundnut was specified. For rice and wheat, harvest was specified at maturity. As the crop production in the study region was under fully irrigated condition, irrigation was set as automatic when required.

3 Results

3.1 Projected changes in maximum temperature (Tmax), minimum temperature (Tmin), and rainfall

A comparison of mean monthly rainfall, Tmax, and Tmin during baseline with that of future scenarios under study helps to understand the magnitude of projected change in these parameters, which will have an influence on the crop growth and yield. It also provides an insight into the variability in projected rainfall, Tmax, and Tmin among 29 GCMs during mid-and end-century periods under RCP4.5 and RCP8.5, when presented using boxplots. Since the cropping system under study covered three seasons in a year, all the months were considered for this study. Mean monthly values of rainfall and Tmax and Tmin were presented in the form of boxplot and the corresponding values during baseline as 'red triangle'. The results are presented in Figs. 1, 2, and 3.

The mean monthly rainfall during the baseline period was almost equal in magnitude to that of the median of 29 GCMs during all the four study periods (combination of mid- and end-century; RCP4.5 and 8.5), except during June. As the amount of rainfall during June to October was higher due to southwest monsoon, the variability among projected rainfall by 29 GCMs was also high, compared to other months (Fig. 1). The highest variability was observed during end-century under RCP8.5. As evident from Fig. 2, the projected Tmax was higher in all months across the GCMs and study periods, except for some outliers lower than that of baseline

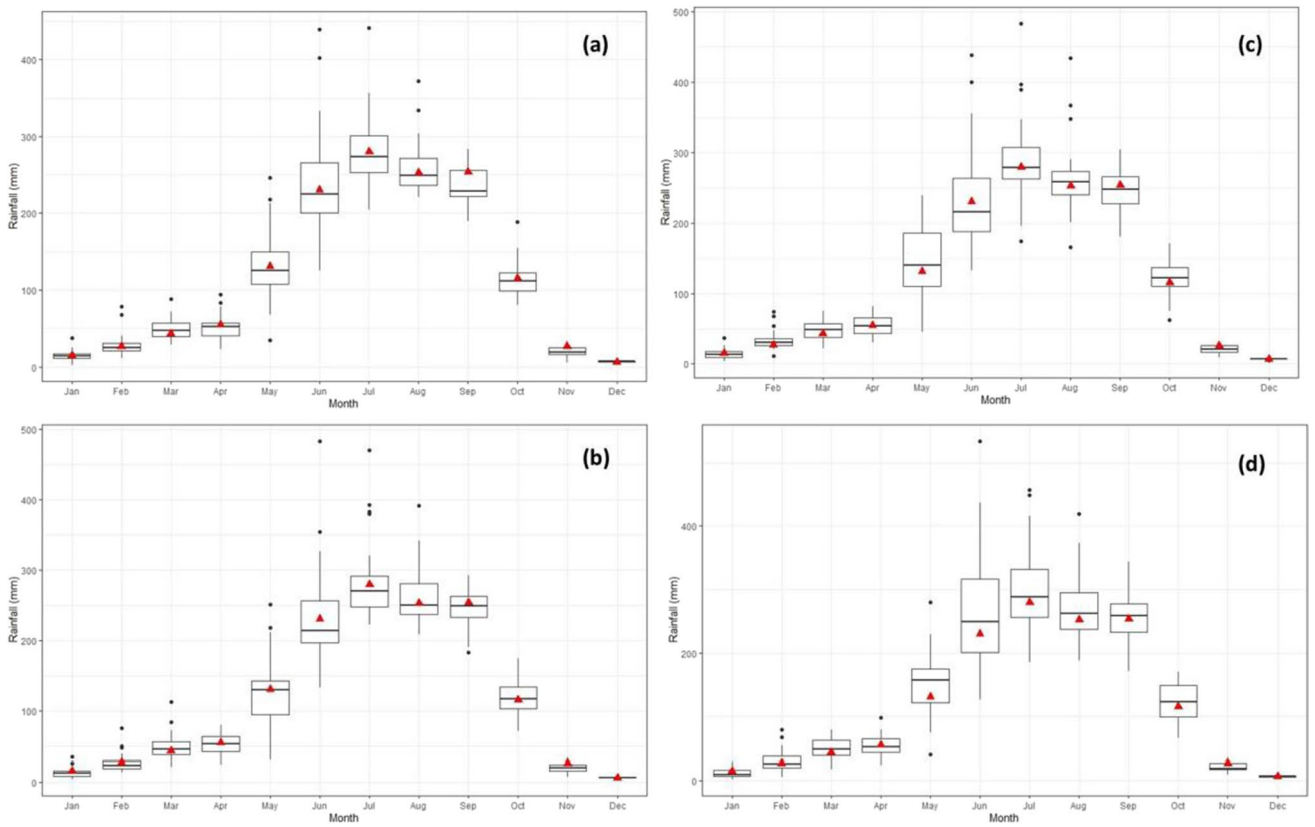


Fig. 1 Comparison of baseline and future rainfall scenarios for Mohanpur, India. Mean monthly rainfall for baseline (1980–2010) is represented as red triangle and 29 GCM scenarios for mid-century-RCP4.5 (**a**), midcentury-RCP8.5 (**b**), end-century-RCP4.5 (**c**), and end-century-RCP8.5 (**d**) are shown as boxplots

during all the months. This difference was higher during end-century under RCP8.5. The result for Tmin was similar to that of Tmax, but the difference between baselines and projected scenarios was higher during the winter months. The highest difference was recorded for end-century, under the RCP8.5 scenario (Fig. 3).

A summary of the projected average crop seasonal rainfall, Tmax, and Tmin compared to the baseline (average of 29 GCMs minus average of baseline) is presented in Table 3. A substantial increase in mean seasonal rainfall (49.1 mm and 60 mm) is projected during the end-century-RCP8.5 scenario for rice and groundnut. An increase in mean seasonal Tmax of more than 2 °C is projected under all scenarios except during mid-century-RCP4.5 (for all crops) and mid-century-RCP8.5 (for rice). More than 3 °C of Tmax is projected during the end-century-RCP8.5 for all crops. Magnitude of increase in Tmin was higher than that of Tmax for all the crops under all scenarios. Table 3 represents the mean changes in rainfall, Tmax, and Tmin during the growing season of rice, wheat, and groundnut, which is projected by GCMs. However, it should be kept in mind that extreme values (towards both higher side and lower side) were present in monthly values of these variables in the form of outliers in Figs. 1, 2, and 3.

3.2 Calibration and validation of crop models

3.2.1 Calibration and validation of CERES-Rice model for cultivar 'Shatabdi'

Calibration provided a set of genetic coefficients for rice cultivar 'Shatabdi' (Supplementary Table 3). The average days taken to anthesis and maturity were 79 and 103 days, respectively. The calibrated model could predict days to anthesis with an RMSE of 3.7, nRMSE of 4.7, and MAE of 3 and days to maturity with an RMSE of 4.2, nRMSE of 4.1, and MAE of 3.6 (Table 4). Similarly, the nRMSE for tops weight and yield was 19.6% (good) and 26% (fair), respectively, which was slightly higher.

3.2.2 Calibration and validation of CERES-Wheat model for cultivar 'PBW-343'

The genetic coefficients optimized for cultivar PBW-343 are presented in Supplementary Table 4. The calibrated model could predict days to anthesis and maturity with an RMSE of 0.6 and 1.4, respectively (Table 4). The nRMSE for the same was 0.7 and 1.1, respectively, which was excellent as

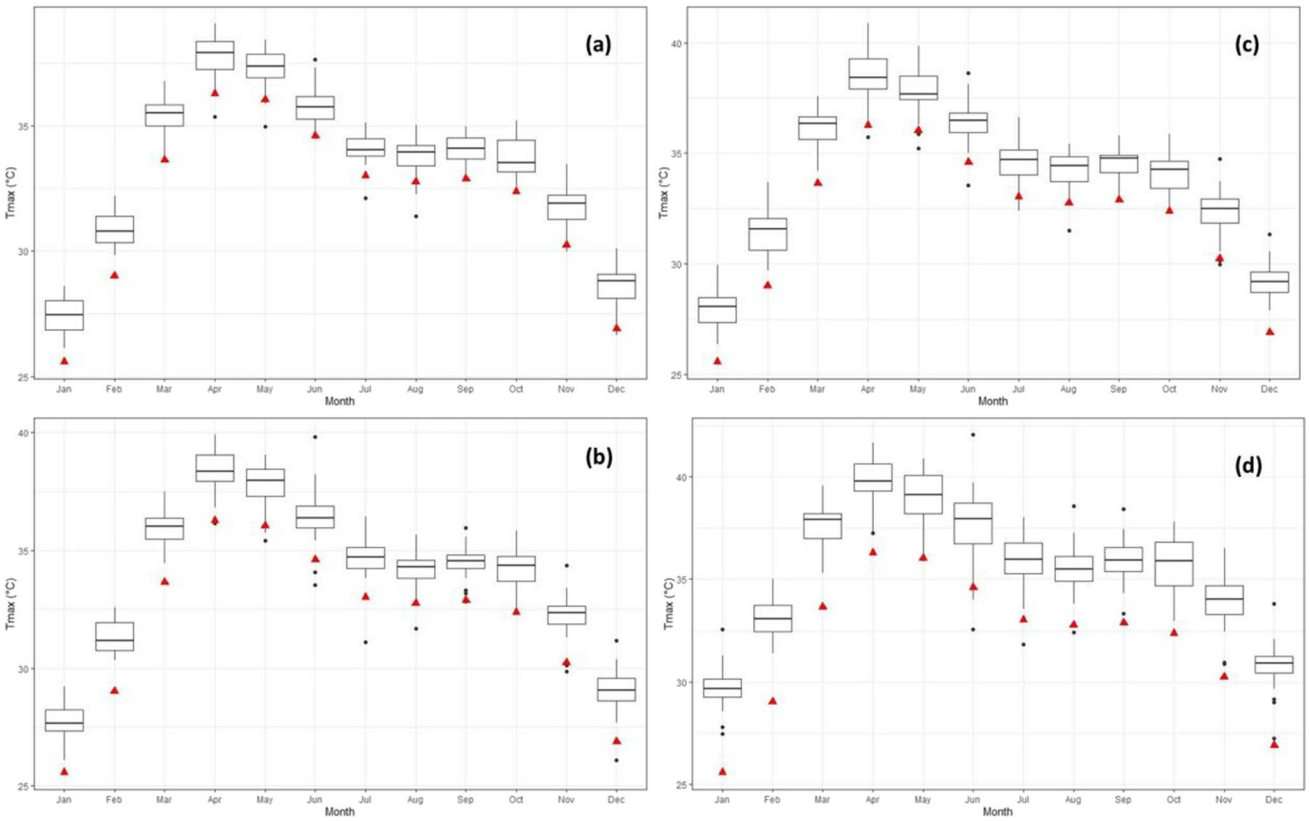


Fig. 2 Comparison of baseline and future maximum temperature (Tmax) scenarios for Mohanpur, India. Mean monthly Tmax for baseline (1980–2010) is represented as red triangle and 29 GCM scenarios for mid-century-RCP4.5 (**a**), mid-century-RCP8.5 (**b**), end-century-RCP4.5 (**c**), and end-century-RCP8.5 (**d**) are shown as boxplots

per Jamieson et al. (1991). In the case of tops weight, RMSE was 1394 kg/ha and nRMSE was 21.1%, which was 'fair'. The model could predict the yield with an RMSE of 457 kg/ha and nRMSE of 19, which was 'good'.

3.2.3 Calibration and validation of CROPGRO-Peanut model for groundnut cultivar 'TG-51'

The genetic coefficients optimized for cultivar TG-51 are presented in Supplementary Table 5. The simulated phenology (days to anthesis and maturity) was well within the acceptable limits with RMSE of 1.3 and 3.3 days and nRMSE of 4.3% and 3.5%, respectively (Table 4). The model could predict tops weight and yield with an nRMSE of 19.3% and 15.4%, respectively.

3.3 Impact of projected climate under RCP4.5 and 8.5 scenarios on phenology, biomass, and yield of rice–wheat-groundnut during mid and end century

This section presents the results of crop sequence analysis done for three time periods, viz., baseline (1980–2010), mid-century (2040–2069), and end-century (2070–2099) under two RCP pathways (4.5 and 8.5) using DSSAT.

3.3.1 Effect of projected climate on crop phenology

The impact of projected climate on phenology was prominent for rice and wheat. We could not find any influence of CO_2 fertilization on crop phenology. There was a substantial reduction in crop duration in terms of days to anthesis and days to maturity under all four future scenarios. The days to anthesis of rice declined by 2–7, 5–8, 5–9, and 9–13 days during mid-century-RCP4.5, mid-century-RCP8.5, end-century-RCP4.5, and end-century-RCP8.5, respectively, when compared to the baseline period. The range indicates the variation in reduction of days to anthesis by five GCMs under the projected climate as shown in Fig. 4. The highest decline of 13 days was simulated with HadGEM2-ES during end-century with RCP8.5 and the lowest decline of 3 days was by MRI-CGCM3 during mid-century-RCP4.5. The days to maturity of rice declined by 4–10, 7–12, 8–13, and 13–17 days under mid-century-RCP4.5, mid-century-RCP8.5, end-century-RCP4.5, and end-century-RCP8.5, respectively. HadGEM2-ES model

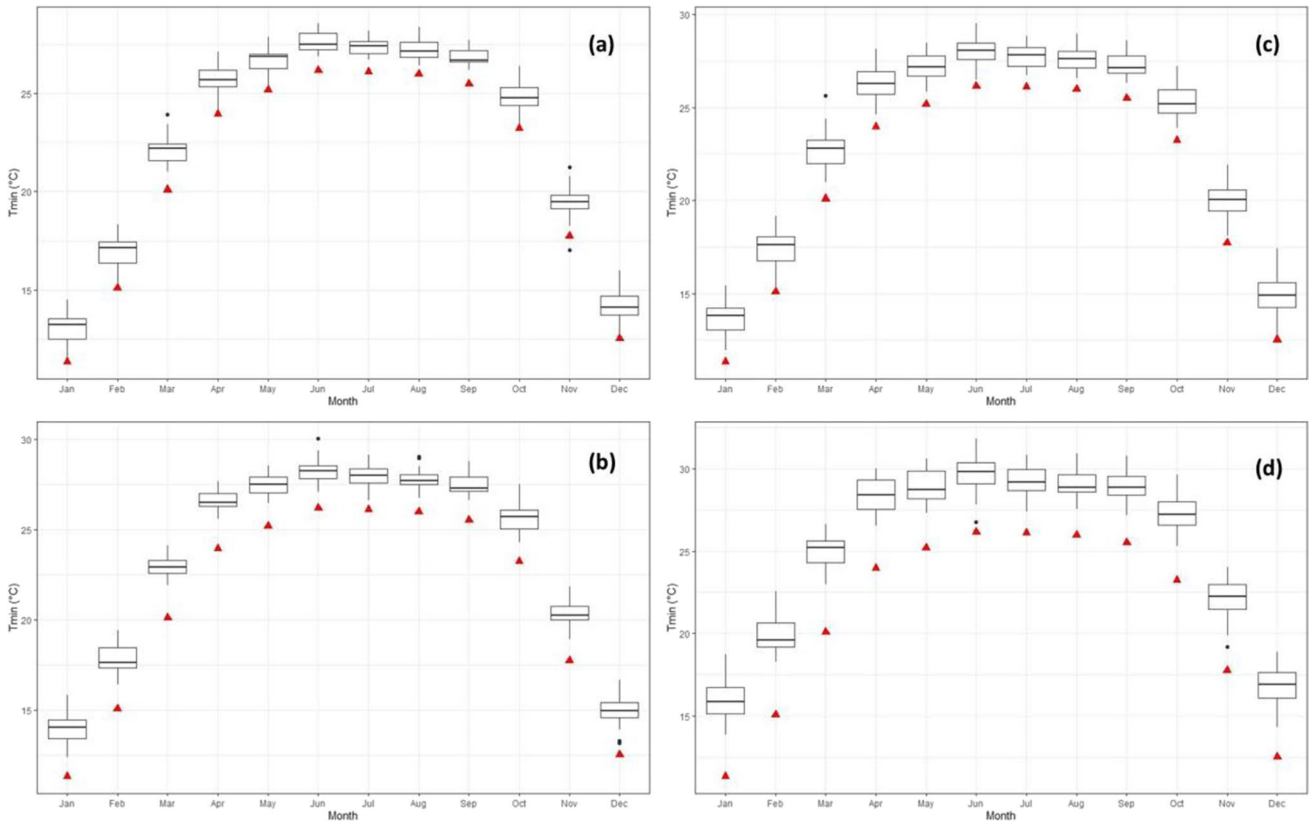


Fig. 3 Comparison of baseline and future minimum temperature (Tmin) scenarios for Mohanpur, India. Mean monthly Tmin for baseline (1980–2010) is represented as red triangle and 29 GCM scenarios for mid-century-RCP4.5 (**a**), mid-century-RCP8.5 (**b**), end-century-RCP4.5 (**c**), and end-century-RCP8.5 (**d**) are shown as boxplots

Table 3 Change in average crop seasonal rainfall, Tmax, and Tmin during future scenarios compared to the baseline at Mohanpur

Variable	Crop	Mid-century-RCP4.5	Mid-century-RCP8.5	End-century-RCP4.5	End-century-RCP8.5
Rainfall (mm)	Rice	−13.2	12.0	21.2	49.1
	Wheat	0.8	−0.1	6.1	1.4
	Groundnut	15.7	2.1	21.0	60.0
Tmax (°C)	Rice	1.1	1.6	1.7	3.0
	Wheat	1.7	2.2	2.3	3.9
	Groundnut	1.4	2.0	2.1	3.4
Tmin (°C)	Rice	1.4	2.0	1.8	3.4
	Wheat	1.8	2.7	2.4	4.6
	Groundnut	1.7	2.5	2.2	4.2

simulated the highest reduction in number of days to maturity (17 days) under end-century-RCP8.5 and MRI-CGCM3 recorded the lowest reduction (4 days). The variation among the GCMs in reduction of days to anthesis and maturity provides a measure of the uncertainty in simulating the impact of projected climate on crop phenology. It was also observed that the reduction in days to anthesis and maturity was more during the RCP8.5 scenario compared to RCP4.5 during end-century compared to that of mid-century.

A similar result was observed for wheat also. The reduction in days to anthesis was 3–7, 5–8, 5–9, and 11–13 days under mid-century-RCP4.5, mid-century-RCP8.5, end-century-RCP4.5, and end-century-RCP8.5, respectively

Table 4 Model evaluation metrics for rice, wheat, and groundnut at Mohanpur

Crops	Variables	Obs	Sim	RMSE	nRMSE	MAE
Rice	Days to anthesis	79	81	3.7	4.7	3
	Days to maturity	103	103	4.2	4.1	3.6
	Tops weight (kg ha^{-1})	7604	7470	1492	19.6	1345
	Yield (kg ha^{-1})	4220	4273	1099	26.0	960
Wheat	Days to anthesis	91	91	0.65	0.7	0.42
	Days to maturity	131	130	1.44	1.1	1.08
	Tops weight (kg ha^{-1})	6603	7835	1394	21.1	1231
	Yield (kg ha^{-1})	2402	2767	457	19.0	374
Groundnut	Days to anthesis	30	30	1.3	4.3	1
	Days to maturity	95	93	3.3	3.5	2.5
	Tops weight (kg ha^{-1})	9051	7448	1750	19.3	1602
	Yield (kg ha^{-1})	2718	2697	418	15.4	408

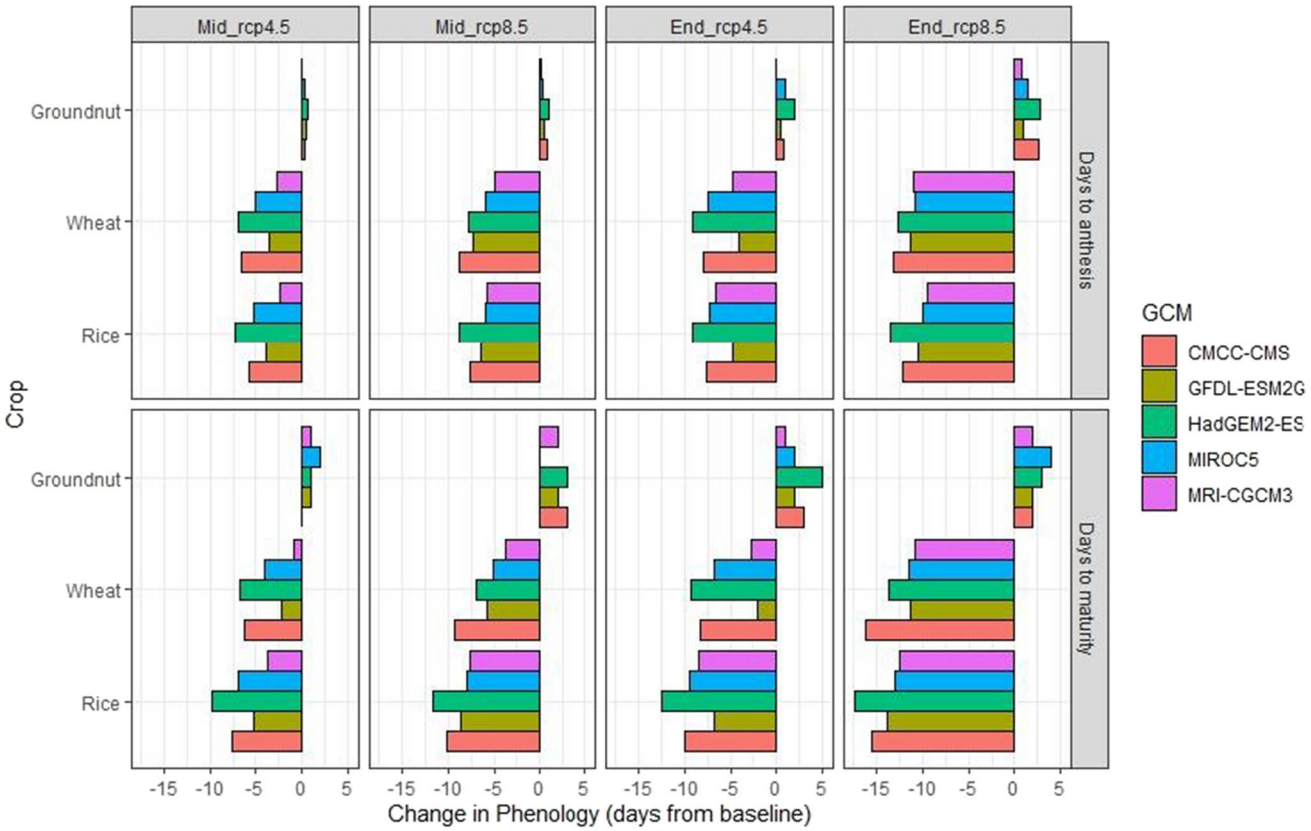


Fig. 4 Impact of projected climate on phenology (changes in days to anthesis and maturity compared to the baseline period) of rice–wheat-groundnut cropping sequence during mid- and end-century conditions and RCP4.5 and RCP8.5 emission scenarios at Mohanpur

(Fig. 4). In the case of days to maturity, the simulated decline in days was 1–6, 4–9, 2–9, and 11–16 days under mid-century-RCP4.5, mid-century-RCP8.5, end-century-RCP4.5, and end-century-RCP8.5, respectively. The lowest decline was simulated by MRI-CGCM3 (3 days for anthesis and one day for maturity) under mid-century-RCP4.5 and the highest decline by CMCC-CMS (13 days for anthesis and 16 days for maturity) under end-century-RCP8.5.

The result obtained for groundnut was different from that of rice and wheat. An average increase of one day in days to anthesis was observed during mid-century. An increase of 3 days was simulated with HadGEM2-ES and MRI-CGCM3 under end-century-RCP8.5. In the case of days to maturity, the increase in days ranged 0–2, 0–3, 2–5, and 2–4 during mid-century-RCP4.5, mid-century-RCP8.5, end-century-RCP4.5, and end-century-RCP8.5, respectively.

3.3.2 Effect of projected climate on biomass production

Effect during mid-century The response of rice biomass to baseline CO_2 and elevated CO_2 under projected climate scenarios was different. Under future climate and baseline CO_2 (360 ppm), an average reduction of 3% in biomass was observed during mid-century-RCP4.5. Under future climate and CO_2 concentration of 499 ppm, a 13% average increase in biomass compared was simulated during mid-century-RCP4.5 (Fig. 5). The change in biomass ranged from − 9.2 (HadGEM2-ES) to 1% (MIROC5) under 360 ppm CO_2 and 6.2% (HadGEM2-ES) to 17.7% (MIROC5) under 499 ppm. For mid-century-RCP8.5, the change in biomass ranged from − 12.1 (HadGEM2-ES) to − 0.2% (MIROC5) under 360 ppm and 9 (HadGEM2-ES) to 21.7% (MIROC5) under 571 ppm.

Unlike rice, the impact of projected climate on wheat biomass was positive under 360 ppm as well as elevated CO_2 concentrations, except with CMCC-CMS during mid-century-RCP8.5 with 360 ppm. During mid-century-RCP4.5, the change in biomass ranged from 0.2 (HadGEM2-ES) to 14.6% (GFDL-ESM2G) under 360 ppm and from 11.2 (HadGEM2-ES) to 27.4% (GFDL-ESM2G) under 499 ppm. The change in biomass was − 5 (CMCC-CMS) to 4.2% (MRI-CGCM3) under 360 ppm and 8.8 (CMCC-CMS) to 20.4% (MRI-CGCM3) under 571 ppm during mid-century-RCP8.5.

Substantial increase in groundnut biomass was observed with projected climate, especially under elevated CO_2 conditions. The range of change in biomass was 11.8 (GFDL-ESM2G) to 16.5% (MIROC5) under 360 ppm and 34 (GFDL-ESM2G) to 39.2% (MIROC5) under 499 ppm during mid-century-RCP4.5. A similar change was observed during mid-century-RCP8.5, which ranged from 12.5 (HadGEM2-ES) to 16.1% (MIROC5) under 360 ppm and from 40.7 (HadGEM2-ES) to 45.2% (MIROC5) under 571 ppm.

Effect during end-century During end-century-RCP4.5, the change in rice biomass ranged from − 21.3 (HadGEM2-ES) to − 0.7% (MIROC5) under 360 ppm and from − 5.6 (HadGEM2-ES) to 17.6% (MIROC5) under 532 ppm (Fig. 6). The highest reduction in rice biomass was observed during end-century-RCP8.5, which varied from − 33.3 (HadGEM2-ES) to − 13% (MIROC5) under 360 ppm. The change was − 3.7 (HadGEM2-ES) to 19.4% (MIROC5) when the CO_2 concentration was 801 ppm.

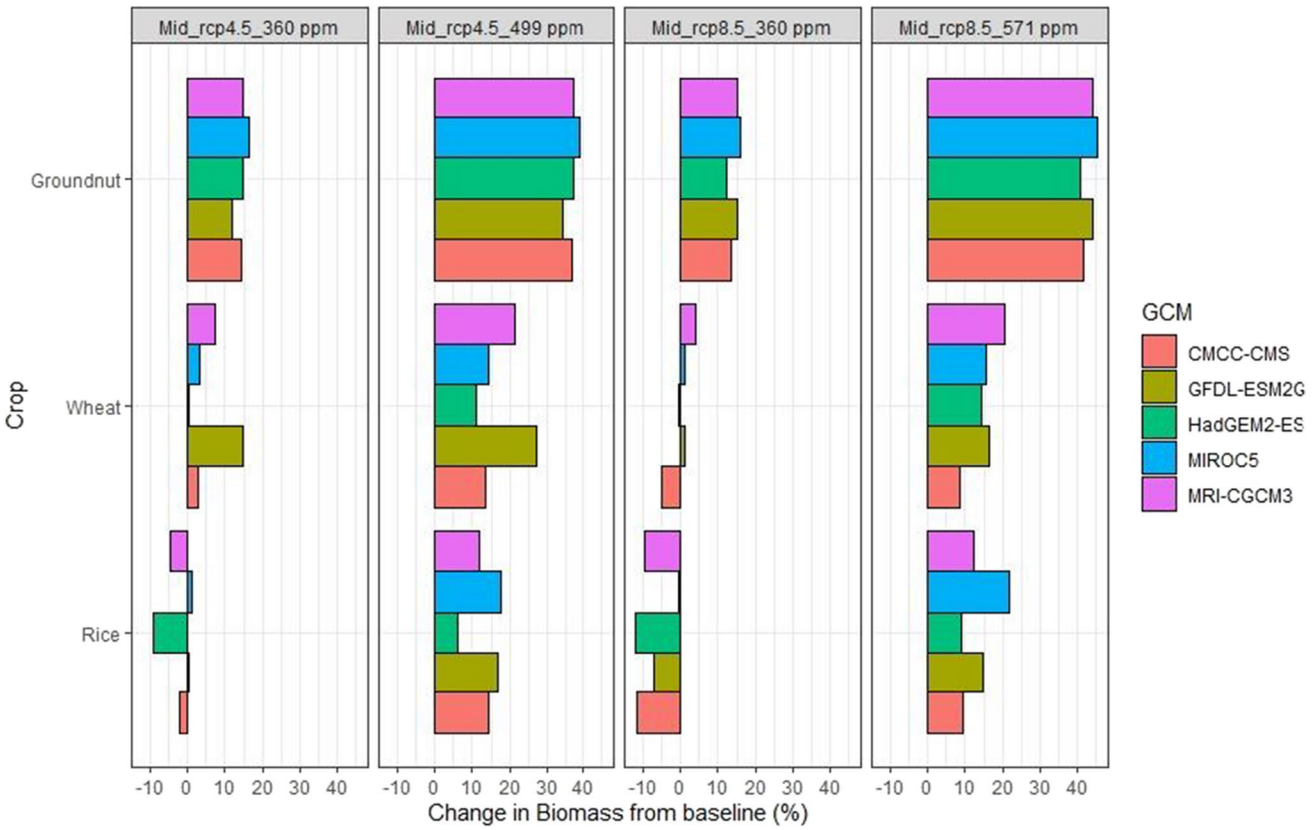


Fig. 5 Impact of projected climate on final biomass (percent change compared to the baseline period) of rice–wheat-groundnut cropping sequence during mid-century under RCP4.5 and RCP8.5 emission scenarios and baseline and elevated CO_2 concentrations at Mohanpur

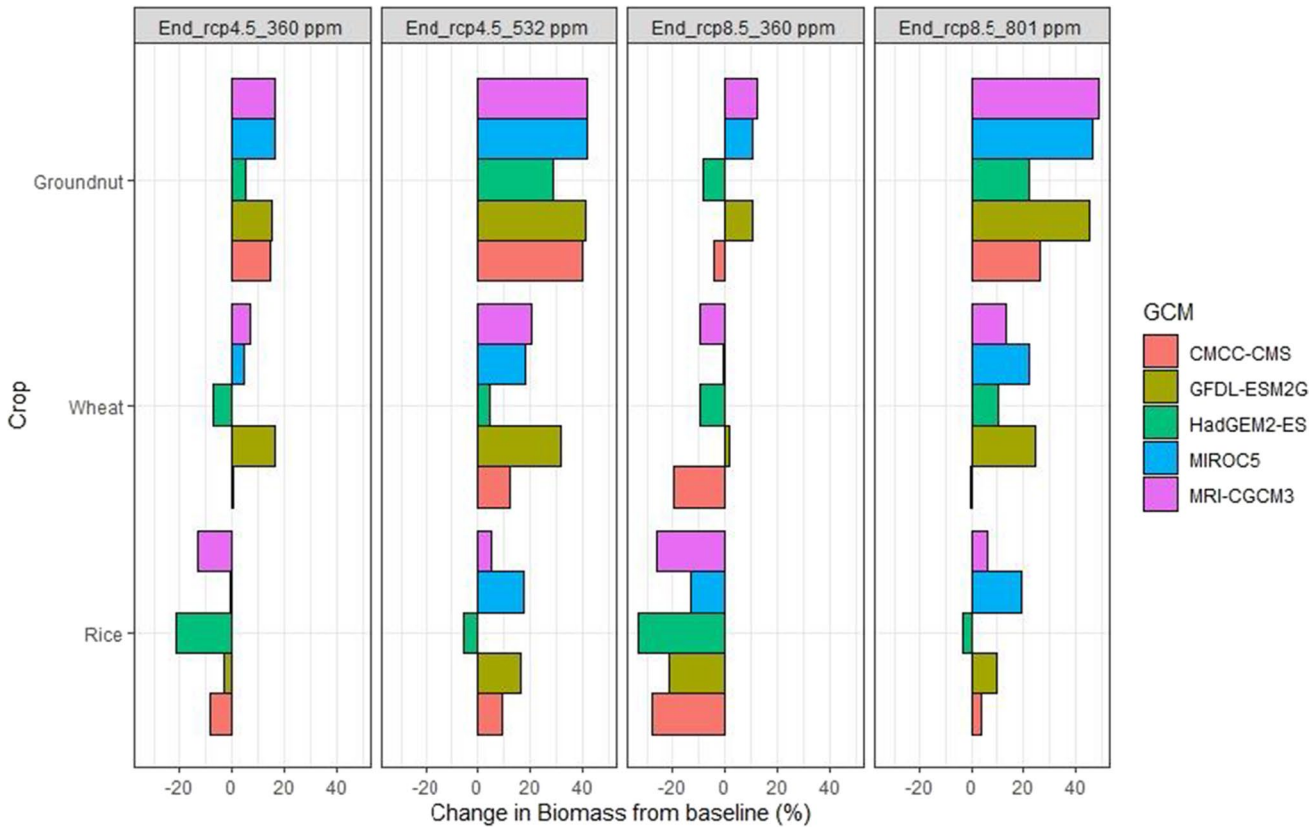


Fig. 6 Impact of projected climate on final biomass (percent change compared to the baseline period) of rice–wheat-groundnut cropping sequence during end-century under RCP4.5 and RCP8.5 emission scenarios and baseline and elevated CO_2 concentrations at Mohanpur

For wheat, during end-century-RCP4.5, the change in biomass varied from − 7.5 (HadGEM2-ES) to 16.3% (GFDL-ESM2G) under 360 ppm and from 4.6 (HadGEM2-ES) to 31.5% (GFDL-ESM2G) under 532 ppm. During end-century-RCP8.5, the value ranged from − 19.7 (CMCC-CMS) to 1.8% (GFDL-ESM2G) when CO_2 concentration was 360 ppm and from − 1 (CMCC-CMS) to 24.8% (GFDL-ESM2G) under 801 ppm.

Change in biomass of groundnut during end-century was positive under all future scenarios and CO_2 concentrations, except for HadGEM2-ES and MRI-CGCM3 during end-century-RCP8.5 with 360 ppm. The highest increase in biomass by 42.1% and 48.7% was recorded with the MRI-CGCM3 model, respectively, under 532 ppm during end-century-RCP4.5 and under 801 ppm during end-century-RCP8.5, respectively.

3.3.3 Effect of projected climate on crop yield

Boxplots were used to show the inter-annual variability in yield within a GCM, variability in projected yield among GCMs and for a comparison with the baseline yield (Figs. 7 and 8). The percent changes in yield are presented in Supplementary Tables 6 to 9.

Mid-century-RCP4.5 scenario The results for mid-century-RCP4.5 under 360 ppm and 499 ppm are presented in Fig. 7 and Supplementary Table 6. Under 360 ppm CO_2 concentration, while three GCMs simulated an increase in rice yield (CMCC-CMS, GFDL-ESM2G, and MIROC5) in the range of 3.6–5.8%, two GCMs simulated a slight decrease in yield (HadGEM2-ES (− 1.8%) and MRI-CGCM3 (− 4.9%)) (Fig. 7a). But the positive impact of elevated CO_2 on yield was observed under 499 ppm, where all the five GCMs simulated a substantial increase in yield and the range was 12.4 (MRI-CGCM3) to 22.9% (MIROC5) (Fig. 7b). The reduction in yield was higher for wheat under 360 ppm and all the GCMs simulated a decrease in yield from baseline, which ranged − 15.1 (GFDL-ESM2G) to − 20.2% (HadGEM2-ES). The magnitude of decrease was reduced under 499 ppm and it ranged between − 5.8 (GFDL-ESM2G) and − 12% (HadGEM2-ES). The results of groundnut showed large variability in a change of yield from baseline among GCMs. Under 360 ppm, only the GFDL-ESM2G model simulated an increase in yield of 3.8%, whereas the other 4 GCMs simulated a decrease in yield which ranged between − 6.5 (MRI-CGCM3) and − 39.8% (HadGEM2-ES). Two GCMs recorded an increase in yield (GFDL-ESM2G (25%) and MRI-CGCM3 (12.3%)) and three GCMs recorded a decrease

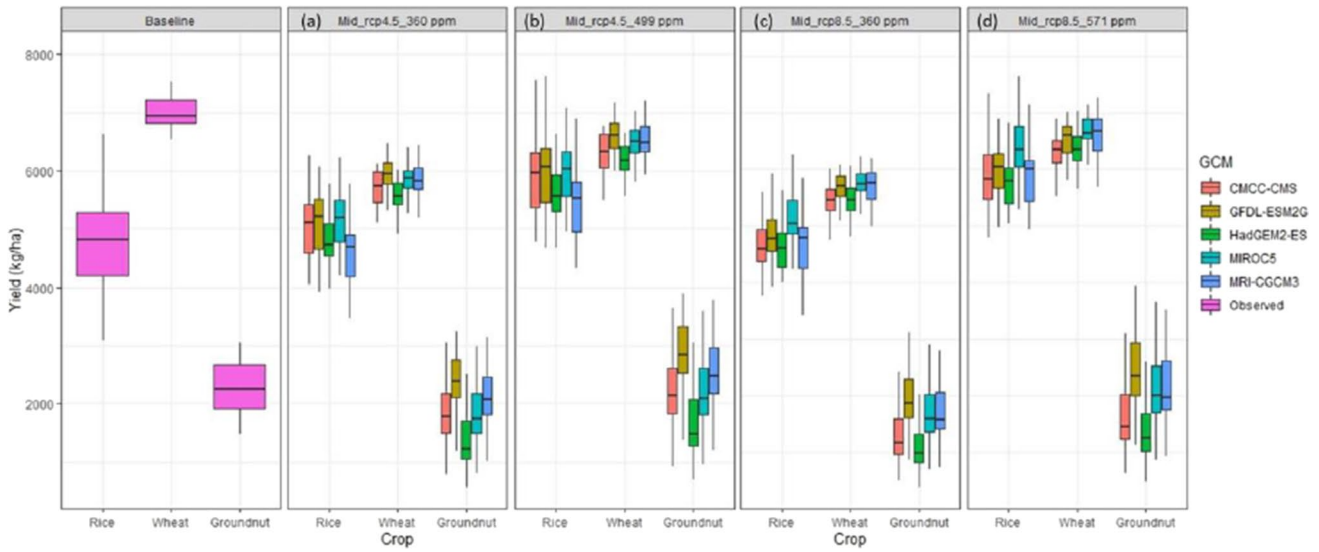


Fig. 7 Impact of projected climate on yield of rice–wheat-groundnut cropping sequence under mid-century-RCP4.5 with baseline CO_2 (**a**), mid-century-RCP4.5 with projected CO_2 (**b**), mid-century-RCP8.5 with baseline CO_2 (**c**), and mid-century-RCP8.5 with projected CO_2 (**d**) at Mohanpur

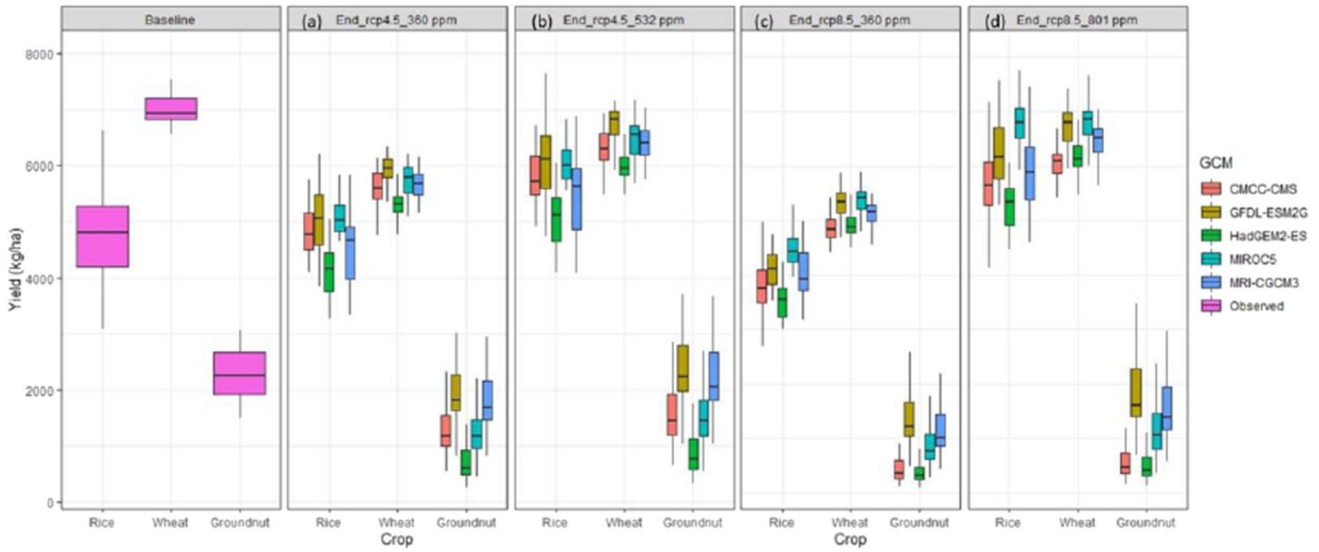


Fig. 8 Impact of projected climate on yield of rice–wheat-groundnut cropping sequence under end-century-RCP4.5 with baseline CO_2 (**a**), end-century-RCP4.5 with projected CO_2 (**b**), end-century-RCP8.5 with baseline CO_2 (**c**), and end-century-RCP8.5 with projected CO_2 (**d**) at Mohanpur

(ranged − 1.8 to − 27.2%) during the simulations under higher CO_2 concentration of 499 ppm.

Mid-century-RCP8.5 scenario The variability in rice yield remained within ± 6% when simulations were performed under 360 ppm CO_2 concentration (Supplementary Table 7). The highest yield reduction of − 5.4% was simulated by HadGEM2-ES and the highest increase of 5.3% was by MIROC5. All the GCMs simulated a substantial increase in yield under 571 ppm (Fig. 7c), which ranged between 19.5 (CMCC-CMS and MRI-CGCM3) and 31.6% (MIROC5). For wheat, the yield reduction under 360 ppm ranged from − 17.8 (MIROC5) to − 21.9% (CMCC-CMS), which was higher compared to the rice. But the elevated CO_2 concentration of 571 ppm could reduce the magnitude of yield reduction in wheat (− 4.9 (MIROC5) to − 9.9% (CMCC-CMS)) (Fig. 7d). Groundnut recorded the highest reduction in yield under 360 ppm and the range of yield reduction was

from – 14.7 (GFDL-ESM2G) to – 52.2% (HadGEM2-ES). The elevated CO_2 of 571 ppm could decrease the magnitude of the yield reduction and the change in yield ranged between 7.5 (GFDL-ESM2G) and – 39.5% (HadGEM2-ES).

End-century-RCP4.5 scenario The inter-annual variability in yield was higher for rice and groundnut compared to wheat which can be observed with the width of box-plots in Fig. 8. Under 360 ppm CO_2 concentration, three GCMs simulated a reduction in rice yield (– 1.4 (CMCC-CMS) to – 14.8% (HadGEM2-ES)) and two GCMs simulated an increase (4.1% by GFDL-ESM2G and 5% by MIROC5) (Fig. 8a, Supplementary Table 8). Simulation under 532 ppm enhanced the rice yield and it ranged from 3.7 (HadGEM2-ES) to 25.9% (MIROC5%) (Fig. 8b). As observed in the earlier scenarios, the yield reduction in wheat was the highest under 360 ppm which ranged from – 15.3 (GFDL-ESM2G) to – 23.9% (HadGEM2-ES). The magnitude of yield reduction was decreased under a CO_2 concentration of 532, for which the range was – 4 (GFDL-ESM2G) to – 14.6% (HadGEM2-ES). Groundnut recorded yield reduction to the tune of – 16.8 (GFDL-ESM2G) to – 68.9% (HadGEM2-ES) under 360 ppm. The elevated CO_2 concentration of 532 ppm reduced the magnitude of yield reduction and it ranged between + 2.3 (GFDL-ESM2G) and – 61.5% (HadGEM2-ES).

End-century-RCP8.5 scenario As this scenario is characterized by a marked increase in temperature (both Tmax and Tmin), the magnitude of yield reduction was more under CO_2 concentration of 360 ppm (Fig. 8c). In the case of rice, the yield reduction was in the range of – 6.9 (MIROC5) to – 26.9% (HadGEM2-ES), which was the highest among all scenarios (Supplementary Table 9). But, under 801 ppm, the impact of CO_2 fertilization was so significant that all the five GCMs simulated an increase in yield in the range of 10.2 (HadGEM2-ES) to 40.9% (MIROC5) (Fig. 8d, Supplementary Table 9). For wheat, the reduction in yield was – 23 (MIROC5) to – 30.5% (CMCC-CMS) under 360 ppm. This was improved under 801 ppm, where the reduction in yield ranged between – 3 (MIROC5) and – 12.8% (CMCC-CMS). Groundnut recorded the highest yield reduction, among the three crops, in the range of – 40.7 (GFDL-ESM2G) to – 84.2% (HadGEM2-ES). Under elevated CO_2 of 801 ppm, the change in yield ranged from – 20.3 (GFDL-ESM2G) to – 78.4% (HadGEM2-ES).

4 Discussion

The variability in results of phenology, above-ground biomass, and yield for rice, wheat, and groundnut crops can be attributed to four sources, viz., (a) variation in projected weather elements by GCMs, (b) differences in working principles of CERES and CROPGRO models, (c) response of the crop models to baseline CO_2 (360 ppm), and elevated CO_2 concentrations in future climate (d) interaction effect of increased Tmax/Tmin, altered rainfall pattern and elevated CO_2 in the future scenarios.

4.1 Future climate trends

Substantial changes in mean annual rainfall were noted during the end-century-RCP8.5 scenario (+ 104 mm) and end-century-RCP4.5 scenario (+ 43.5 mm). The projected increase in mean annual Tmax was in the range of 1.4 (mid-century-RCP4.5) to 3.4 °C (end-century-RCP8.5). In the case of Tmin, the range was from 1.6 (mid-century-RCP4.5) to 4.1 °C (end-century-RCP8.5). According to the IPCC report (2007), Gangetic West Bengal would experience less rain and a 0.5–1 °C rise in average temperature between 2020 and 2029, and a 3.5–4.5 °C rise between 2030 and 2099. In the case of rainfall, Ghosh (2018) found a significant reduction for Gangetic West Bengal in the early monsoonal month of June and the mid monsoonal month of August. The same result was observed in the present study during mid-century under RCP8.5 and during end-century under RCP4.5.

4.2 Impact of future climate on phenology

There was a marked reduction in days to anthesis and maturity for rice and wheat under future climate compared to baseline. The reduction was more under RCP8.5 than RCP4.5. Rising temperatures (both Tmax and Tmin) cause shortening of crop growth duration and thereby reduces the yield (Zhang et al., 2017). The range of reduction in days to anthesis for rice was 2–13 days and days to maturity were 4 to 17 days (from mid-century-RCP4.5 to end-century-RCP8.5). This can be attributed to the increase in Tmax, Tmin during the crop growth season. During the mid-century-RCP4.5 scenario, an increase in mean seasonal Tmax and Tmin of 1.1 and 1.4 °C respectively has caused 5 and 7 days reduction to reach the anthesis and maturity stage, respectively, for rice. A similar reduction in days to anthesis in the range of 2.4 to 9.5 days under RCP4.5 and 2.3 to 14.9 days under RCP8.5 was reported by Li et al. (2016) in Hunan province, China, during 2080s. Under the same scenario, wheat attained anthesis and maturity 5 and 4 days, early due to an increase in Tmax (by 1.7 °C) and Tmin (1.8 °C), respectively. Under the extreme-scenario (end-century-RCP8.5), the duration of rice was shortened by 14 days with an increase in Tmax and Tmin by 3 and 3.4 °C, respectively. For wheat, the duration reduced by 13 days with an increase in Tmax and Tmin by 3.9 °C and 4.6 °C, respectively. Wang et al. (2015) had reported an average advancement in days to anthesis in wheat by 10.2 days

for RCP4.5 and 17.8 days for RCP8.5 across the New-South Wales wheat belt in eastern Australia during 2061–2100. A reduction in duration from flowering to maturity in the range of 2.2 to 8.9 days in spring wheat across five locations in China during 2071–2100 was reported by Xiao et al. (2018). But, surprisingly for groundnut, a marked reduction was not observed in either days to anthesis or maturity. On the contrary, during end-century-RCP8.5 scenario, the days to anthesis and maturity increased by 2 and 3 days, respectively, in response to an increase in Tmax and Tmin by 3.4 °C and 4.2 °C, respectively. One possible reason for this may be the increase in projected summer rainfall, especially during end_rcp8.5, where an increase in more than 60-mm rainfall is expected compared to the baseline average. According to a simulation study conducted by Singh et al. (2008), as the temperature rises, rain becomes the deciding factor in crop production regulation. The increase in temperature, if any, is likely to be balanced by an increase in rainfall.

4.3 Impact of future climate on above-ground biomass

A reduction in biomass is expected as a result of reduction in crop duration and higher photorespiration due to increase in average seasonal Tmax and Tmin. But the magnitude of this reduction is lessened by the CO_2 fertilization effect. Higher CO_2 concentration in the atmosphere will increase the rate of photosynthesis and higher biomass accumulation in crops. The net rate of biomass accumulation will be based on the interaction between higher temperature and enhanced CO_2. In the case of rice, this concept was observed, where a reduction in biomass was recorded under all the four scenarios with baseline CO_2 concentration (360 ppm). From this, it is clear that the negative impact of elevated temperature was higher compared to positive impact of baseline CO_2. But, under elevated CO_2 condition during all the future scenarios, an increase in accumulated rice biomass was simulated by all the GCMs, except by HadGEM2-ES during end-century-RCP 4.5 and 8.5. The results for wheat were also on similar lines, except the fact that the change in biomass was slightly positive even with baseline CO_2.

Groundnut recorded an increase in biomass when simulated with both baseline and elevated CO_2 under all four future scenarios and GCMs, except by HadGEM2-ES and CMCC-CMS models during end-century-RCP8.5. Under elevated CO_2 concentration, an average increase in biomass of 43.1% and 37.8% was simulated during end-century RCP4.5 and RCP8.5, respectively. Many studies have reported similar results. Prasad et al. (2003) had reported that even over temperatures of 36 °C, the rate of photosynthesis and biomass accumulation in groundnut increased. In the present study, the mean monthly Tmax was ≥ 36 °C during groundnut growing season under all future climate scenarios. They reported an average increase of 36% in total dry matter production across all temperatures. Pilumwong et al. (2007) and Bannayan et al. (2009) reported an increase in groundnut biomass at elevated CO_2 irrespective of the temperature regime. Rajwade et al. (2015) found that biomass increased from baseline to 2020, 2050, and 2080 at their study locations in eastern India.

4.4 Impact of future climate on crop yield

Among the three crops under study, the yield of rice was simulated to increase by 4 to 40% under elevated CO_2 during all the four future scenarios. For wheat, the reduction in yield was in the range of 3 to − 14%, under all the future scenarios under elevated CO_2. Lal et al. (1998) had reported a 15% increase in yield of rice when the CO_2 concentration was doubled from 330 ppm over north-west India. They had also concluded that a 3 °C (2 °C) rise in air temperature nearly cancels out the positive effect of elevated CO_2 on the wheat (rice) yields. Rice being a C_3 crop, responded positively to elevated CO_2. The higher concentration of CO_2 in the future climate has counteracted the negative impact of an increase in temperature, although it could not fully compensate for the adverse impact of high temperature in the case of wheat.

The yield of groundnut was projected to decrease drastically by 25% and 50% during end-century-RCP 4.5 and 8.5, compared to the baseline. One reason for the better performance of rice could be due to the fact that the increase in seasonal average Tmax and Tmin was the least during its growing season compared to wheat and groundnut. Another reason could be the availability of higher residue mass for rice from preceding groundnut crop. Groundnut, being a legume, might have improved soil fertility. The increase in seasonal average Tmax and Tmin was the highest for groundnut, which was 3.4 °C and 4.2 °C during end-century-RCP8.5, which could have adversely affected yield. Prasad et al. (2003) reported a decrease in pollen viability with an increase in temperature under both ambient and elevated CO_2 concentration. They also noted the decrease in seed yield in Groundnut under elevated CO_2 conditions when the temperature was greater than 36 °C, despite greater photosynthesis and biomass accumulation.

4.5 Uncertainty in impact of future climate on crop phenology, biomass, and yield due to choice of GCMs

The difference in impact of projected climate on duration, biomass, and yield by GCMs during the future scenarios were mainly due to the differences in rainfall, Tmax, and Tmin among them, which is presented in Figs. 1, 2, and 3. The reduction in crop duration, biomass, and yield was the

highest by simulation of HadGEM2-ES for the three crops under almost all scenarios because it represented the 'hot/dry' category, where the mean annual temperature is around 2.7 °C more and mean annual rainfall is approximately same as the baseline period. The selected five GCMs represented the whole spectrum of 29 GCMs, hence provide an estimation of the variability in crop duration, biomass, and yield that can be expected by the selection of GCMs.

4.6 Limitations of the study

This study is undertaken using only one crop model, DSSAT, to study the impact of projected climate on rice–wheat-groundnut cropping system. Every crop model will have substantial uncertainty. Some approaches to improve this were reported by Asseng et al. (2013) and Martre et al. (2015) by the use of ensembles of crop models which provided a better prediction of yield than using one crop model.

The CO_2 concentration for the future period was assigned based on the AgMIP central year CO_2 assumption as described in materials and methods. It may not represent the actual increase over the 30 years during mid and end-centuries under RCP4.5 and RCP8.5.

5 Conclusions

A remarkable increase in future mean seasonal maximum and minimum temperature was projected in the study area by multiple GCMs. While the yield of rice was projected to increase, wheat and groundnut yields were likely to decrease during all future scenarios. The projected climate reduced the crop duration in rice and wheat under all future scenarios. The climate change impact analysis revealed a positive impact of CO_2 fertilization on biomass accumulation for rice and wheat under elevated CO_2 concentrations (499 ppm, 571 ppm, 532 ppm, and 801 ppm during mid-century-RCP4.5, mid-century-RCP8.5, end-century-RCP4.5, and end-century-RCP8.5, respectively). However, the effect was negative at 360 ppm of CO_2 concentrations for both the crops. On the other hand, groundnut showed a substantial increase in biomass under all elevated CO_2 concentrations and RCP scenarios. The difference in response of crops in a sequence to the interaction of projected Tmax and Tmin with elevated and baseline CO_2 was revealed in this study. The uncertainty in projected changes of phenology, biomass, and yield due to GCMs was also quantified. Climate change impact analysis of the crop sequence revealed that, although there was a substantial yield reduction of groundnut under elevated CO_2 conditions, the inclusion of groundnut in the cropping sequence has benefitted succeeding rice crop in long-term. This information can be further used for developing crop adaptation strategies to combat the impact of climate change.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00704-021-03700-2.

Acknowledgements This study is part of PhD research work of the first author, who acknowledges the study leave granted by his employer, ICAR-Central Research Institute for Dryland Agriculture, Hyderabad, Telangana.

Author contribution Methodology, Software, writing-original draft, visualization: M. A. Sarath Chandran.
Conceptualization, investigation: Saon Banerjee:
Supervision: Manoj K. Nanda.
Resources: Asis Mukherjee.
Review and editing, result interpretation: V. Visha Kumari.
Review and editing: Subhasis Mondal.

Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Code availability The custom codes used for data visualization in this study will be shared upon request.

Declarations

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Conflict of interest The authors declare no competing interests.

References

AgMIP (2012) Guide for regional integrated assessments: handbook of methods and procedures, Version 4.2. AgMIP, URL: http://www.agmip.org/wp-content/uploads/2013/06/AgMIP-Regional-Research-Team-Handbook-v4.2.pdf

AgMIP (2013) Guide for running AgMIP climate scenario generation tools with R in Windows Version 2.3. http://www.agmip.org/wp-content/uploads/2013/10/Guide-for-Running-AgMIP-Climate-Scenario-Generation-with-R-v2.3.pdf

AICRP on Agrometerology, Mohanpur (2016). Annual report. Bidhan Chandra Krishi Viswavidyalaya, Mohanpur

Asseng S, Ewert F, Rosenzweig C et al (2013) Uncertainty in simulating wheat yields under climate change. Nat Clim Change 3:827–832

Bannayan M, TojoSoler CM, Garcia AGY, Guerra LC, Hoogenboom G (2009) Interactive effects of elevated [CO_2] and temperature on growth and development of a short- and long-season peanut cultivar. Clim Chang 93(3–4):389–406

Bassu S, Brisson N, Durand JL et al (2014) How do various maize crop models vary in their responses to climate change factors. Glob Change Biol 20(7):2301–2320

Banerjee S, Das S, Mukherjee A, Mukherjee A, Saikia B (2016) Adaptation strategies to combat climate change effect on rice and mustard in Eastern India. Mitig Adapt Strateg Glob Change 21:249–261

Ewert F, Rotter RP, Bindi M et al (2015) Crop modelling for integrated assessment of risk to food production from climate change.

Environ Model Software 72:287–303. https://doi.org/10.1016/j.envsoft.2014.12.003

Fangzauva D (2011) Response of Groundnut and Rice crops to different meteorological conditions (Ph.D. thesis). Accessed from the library, Bidhan Chandra Krishi Viswavidyalaya, Mohanpur, Nadia, W.B., India

Ghosh KG (2018) Analysis of rainfall trends and its spatial patterns during the last century over the Gangetic West Bengal. Eastern India J Geovis Sp Analy 2:15. https://doi.org/10.1007/s41651-018-0022-x

Hoogenboom G, Jones JW, Wilkens PW, Porter CH, Batchelor WD, Hunt LA, Boote KJ, Singh U, Uryasev O, Bowen WT, Gijsman AJ, Du Toit A, White JW, Tsuji GY (2004) Decision Support System for Agrotechnology Transfer Version 4.0 (CD-ROM). Honolulu, HI: University of Hawaii

IPCC: Intergovernmental Panel on Climate Change. (2007). Climate change 2007: The physical science basis. In: (eds.) Solomon S, Quin D, Manning M, Chen X, Marquis M, Averyt KB, Tignor HL, Miller M. Contribution of working group I to the fourth assessment report of the intergovernmental panel on climate change, 1–996. Cambridge: Cambridge University Press.

Jamieson PD, Porter JR, Wilson DR (1991) A test of the computer simulation model ARCWHEAT1 on wheat crops grown in New Zealand. Field Crop Res 27:337–350. https://doi.org/10.1016/0378-4290(91)90040-3

Jones JW, Hoogenboom G, Porter CH, Boote KJ, Batchelor WD, Hunt LA, Wilkens PW, Singh U, Gijsman AJ, Ritchie JT (2003) DSSAT cropping system model. Eur J Agron 18:235–265

Jones JW, He J, Boote KJ, Wilkens P, Porter CH, Hu Z (2011) Estimating DSSAT cropping system cultivar-specific parameters using Bayesian techniques. In: Ahuja LR, Ma L (eds.) Methods Introd Syst Model Into Agric Res pp. 365–394. https://doi.org/10.2134/advagricsystmodel2.c13

Kollas C, Kersebaum KC, Nendel C et al (2015) Crop rotation modelling—a European model Intercomparison. Eur J Agron 70:98–111

Lal M, Singh KK, Rathore LS, Srinivasan G, Saseendran SA (1998) Vulnerability of rice and wheat yields in NW India to future changes in climate. Agric for Meteorol 89:101–114

Li Y, Wu W, Ge Q, Zhou Y, Xu C (2016) Simulating climate change impacts and adaptive measures for rice cultivation in Hunan province, China. J Appl Met Clim 55:1359–1376. https://doi.org/10.1175/JAMC-D-15-0213.1

Lin BB (2011) Resilience in agriculture through crop diversification: adaptive management for environmental change. Bioscience 61:183–193

Martre P, Wallach D, Asseng S et al (2015) Multimodel ensembles of wheat growth: many models are better than one. Glob Change Biol 21:911–925. https://doi.org/10.1111/gcb.12768

Mathews RB, Rivington M, Muhammed S, Newton AC, Hallet PD (2013) Adapting crops and cropping systems to future climates to ensure food security: the role of crop modelling. Glob Food Sec 2:24–28

Olesen JE, Bindi EM (2002) Consequences of climate change for European agricultural productivity, land use and policy. Eur J Agron 16:239–262

Olesen JE, Trnka M, Kersebaum KC, Skjelvag AO, Seguin B, Peltonen-Sainio P, Rossi F, Kozyra J, Micale F (2011) Impacts and adaptation of European crop production systems to climate change. Eur J Agron 34(2):96–112. https://doi.org/10.1016/j.eja.2010.11.003

Parton WJ, Scurlock JMO, Ojima DS, Gilmanov TG, Scholes RJ, Schimel DS, Kirchner T, Menaut JC, Seastedt T, Garcia Moya E, Kamnalrut A, Kinyamario JL (1993) Observations and modeling of biomass and soil organic matter dynamics for the grassland biome worldwide. Global Biogeochem Cycles 7:785–809

Palosuo T, Kersebaum KC, Angulo C et al (2011) Simulation of winter wheat yield and its variability in different climates of Europe: a comparison of eight crop growth models. Eur J Agron 35(3):103–114

Pilumwong J, Senthonga C, Srichuwongb S, Ingram KT (2007) Effects of temperature and elevated CO_2 on shoot and root growth of peanut (Arachis hypogaea L.) grown in controlled environment chambers. Sci Asia 33:79–87

Prasad PVV, Boote KJ, Allen LH, Thomas JMG (2003) Super-optimal temperatures are detrimental to peanut (Arachis hypogaea L.) reproductive processes and yield under both ambient and elevated carbon dioxide. Glob Change Biol 9:1775–1787

Rahman MH, Ahmad A, Wang X, Wajid A, Nasim W, Hussain M, Ahmad B, Ahmad I, Ali Z, Ishaque W, Awais M, Shelia V, Ahmad S, Fahd S, Alam M, Ullah H, Hoogenboom G (2018) Multi-model projections of future climate and climate change impacts uncertainty assessment for cotton production in Pakistan. Agric for Meteorol 253–254:94–113

Rajwade YA, Swain DK, Tiwari KN, Mohanty UC, Goswami P (2015) Evaluation of field level adaptation measures under the climate change scenarios in rice based cropping system in India. Environ Process 2:669–687. https://doi.org/10.1007/s40710-015-0115-1

Rötter RP, Palosuo T, Kersebaum KC et al (2012) Simulation of spring barley yield in different climatic zones of Northern and Central Europe: a comparison of nine crop models. Field Crops Res 133:23–36

Ruane AC, Winter JM, McDermid SP, Hudson NI (2015) AgMIP climate data and scenarios for integrated assessment. In: Rosenzweig C, Hillel D (Eds), Handbook of Climate Change and Agroecosystems: The Agricultural Model Intercomparison and Improvement Project (AgMIP). ICP Series on Climate Change Impacts, Adaptation, and Mitigation Vol. 3. Part 1, Imperial College Press, p 45–78, doi: https://doi.org/10.1142/9781783265640_0003

Ruane AC, McDermid SP (2017) Selection of a representative subset of global climate models that captures the profile of regional changes for integrated climate impacts assessment. Earth Persp 4:1. https://doi.org/10.1186/s40322-017-0036-4

Reidsma P, Ewert F, Boogaard H, van Diepen K (2009) Regional crop modelling in Europe: the impact of climatic conditions and farm characteristics on maize yields. Agric Syst 100(1):51–60

Singh M, Kalra N, Chakraborty D, Kamble K, Barman D, Saha S, Mittal RB, Pandey S (2008) Biophysical and socioeconomic characterization of a water-stressed area and simulating agri-production estimates and land use planning under normal and extreme climatic events: a case study. Environ Monit Assess 142:97–108

Smith RG, Gross KL, Robertson GP (2008) Effects of crop diversity on agro ecosystem function: crop yield response. Ecosystems 11:355–366

Soler CMT, Sentelhas PC, Hoogenboom G (2007) Application of the CSM-CERES-maize model for planting date evaluation and yield forecasting for maize grown off-season in a subtropical environment. Eur J Agron 27:165–177. https://doi.org/10.1016/j.eja.2007.03.002

Teixeira EI, Brown HE, Sharp JM, Meenken ED, Ewert F (2015) Evaluating methods to simulate crop rotations for climate impact assessments—a case study on the Canterbury plains of New Zealand. Environ Modell Softw 72:304–313. https://doi.org/10.1016/j.envsoft.2015.05.012

Thentu TL (2016) Evaluation of Wheat productivity response to irrigation and fertlizer in West Bengal using field experiments and crop simulation modelling (PhD thesis). Accessed from the library, Bidhan Chandra Krishi Viswavidyalaya, Mohanpur, Nadia, W.B., India

Timsina J, Singh U, Singh Y (1997) Addressing sustainability of rice-wheat systems: analysis of long-term experimentation

and simulation. In: Kropff MJ et al. (eds.): applications of systems approaches at the field level 383–397. Kluwer Academic Publishers

Tsuji GY, Jones JW, Hoogenboom G, Hunt LA, Thornton PK (1994) Introduction. In: Tsuji GY, Uehara G, Balas S (eds) DSSAT v3, Decision Support System for Agrotechnology Transfer, vol 1. University of Hawaii, Honolulu, HI, pp 1–20

Wang B, Liu DL, Asseng S, Macadam I, Yu Q (2015) Impact of climate change on wheat flowering time in eastern Australia. Agric for Meteorol 209–210:11–21. https://doi.org/10.1016/j.agrformet.2015.04.028

Xiao D, Bai H, Liu DL (2018) Impact of future climate change on wheat production: a simulated case for China's wheat system. Sustainability 10:1227. https://doi.org/10.3390/su10041277

Zhang Q, Zhang W, Li T, Sun W, Yu Y, Wang G (2017) Projective analysis of staple food crop productivity in adaptation to future climate change in China. Int J Biometeorol 61:1445–1460. https://doi.org/10.1007/s00484-017-1322-4

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.